



NPTEL Course on “Computer System Optimization for ML”

Lecture #1



Agenda

- **Timeline**
- **Introduction**

NPTREL

Timeline of the Course (1)

Date	Topic covered
Lecture-1	Introduction
Lecture-2	Convolutional Neural Network (1)
Lecture-3	Convolutional Neural Network (2)
Lecture-4	CNN Accelerators (1)
Lecture-5	CNN Accelerators (2)
Lecture-6	Pruning and Quantization
Lecture-7	Parallelism in ML Training (1)
Lecture-8	Parallelism in ML Training (2)
Lecture-9	Checkpointing in ML Training
Lecture-10	Basics of Large Language Models (LLM)
Lecture-11	Scheduling : Efficient LLM scheduling and routing (1)
Lecture-12	Scheduling : Efficient LLM scheduling and routing (2)

Timeline of the Course (2)

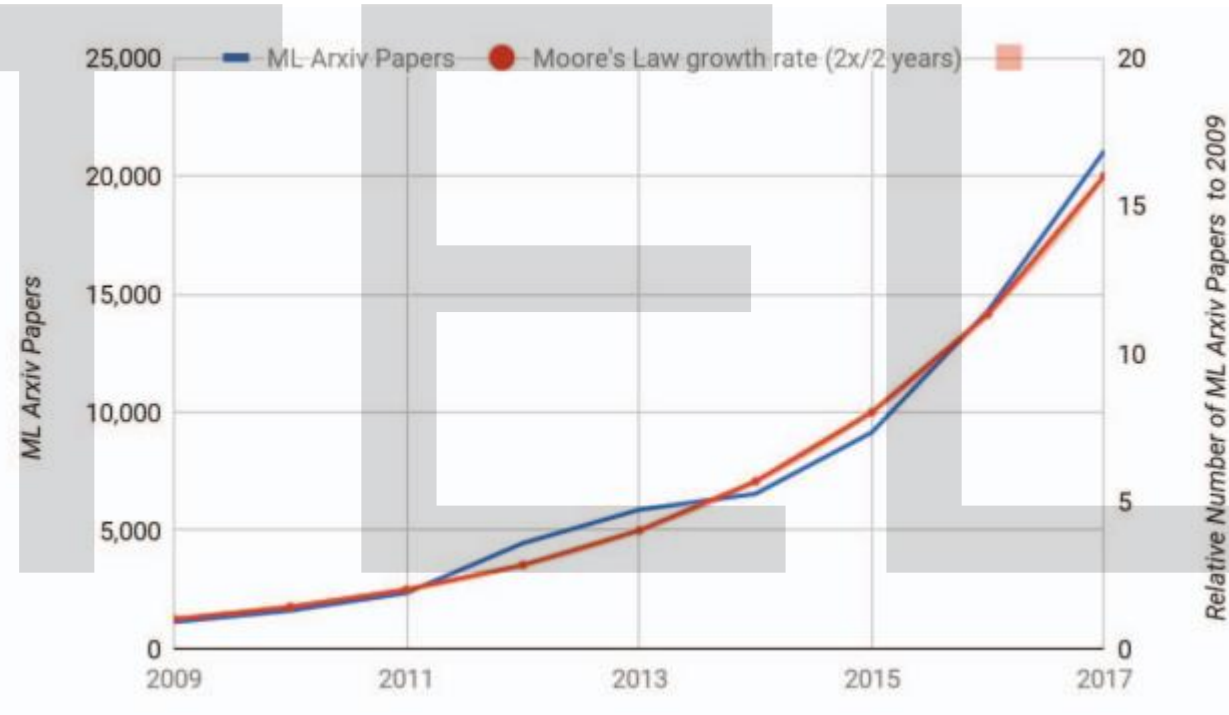
Date	Topic covered
Lecture-12	Scheduling : Efficient LLM scheduling and routing (3)
Lecture-13	Accelerating decoding: Speculative decoding
Lecture-14	Introduction to Graph Neural Network (GNN)
Lecture-15	Accelerators for Graph Convolutional Network (GCN)
Lecture-16	CNN Accelerators with Emerging Technology
Lecture-17	Attention Acceleration with Emerging Technology
Lecture-18	GCN Accelerators with Emerging Technology
Lecture-19	<Blanket Day>
Lecture-20	<Blanket Day>

Key Takeaway from the Course

- **Tools**
 - Latex
 - Weka
 - Origin
- **Communication skill (a key in your engagement after your degree)**
 - Taking notes
 - Preparing reports for the project
 - Project presentation
- **Strong analysis skill**
 - Is useful for daily life 😊
- **Networking**
 - Group project
- **Timeliness**

Why Machine Learning?

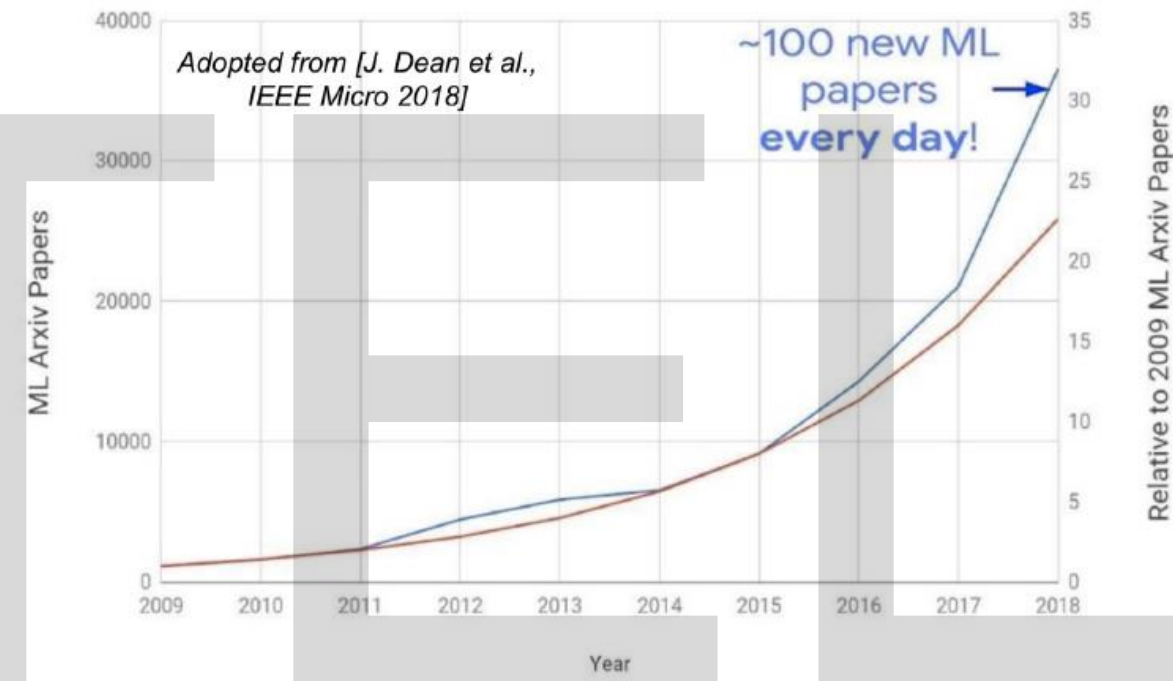
- There exists several estimation/prediction task which have several factors to be considered
 - Weather prediction
 - Face detection
 - Sentence construction
- They are difficult to model analytically
 - e.g., distance = velocity x time is easy
- The exact analytical model are too compute intensive
 - Navier-stokes equation
- Machine learning (ML) provides the scope to learn the models given enough prior observations



Adopted from J. Dean et al., IEEE Micro 2018

We have Reached the Third Wave

- The third wave is here
- ML's meteoric raise
- After 2 hype-disillusionment cycles, the third wave has crossed the threshold
- And it is further increasing
 - **13,300** papers submitted in NeurIPS'23
 - **3540** were accepted
 - Check the data yourself for AAAI, ICLR, ICML etc

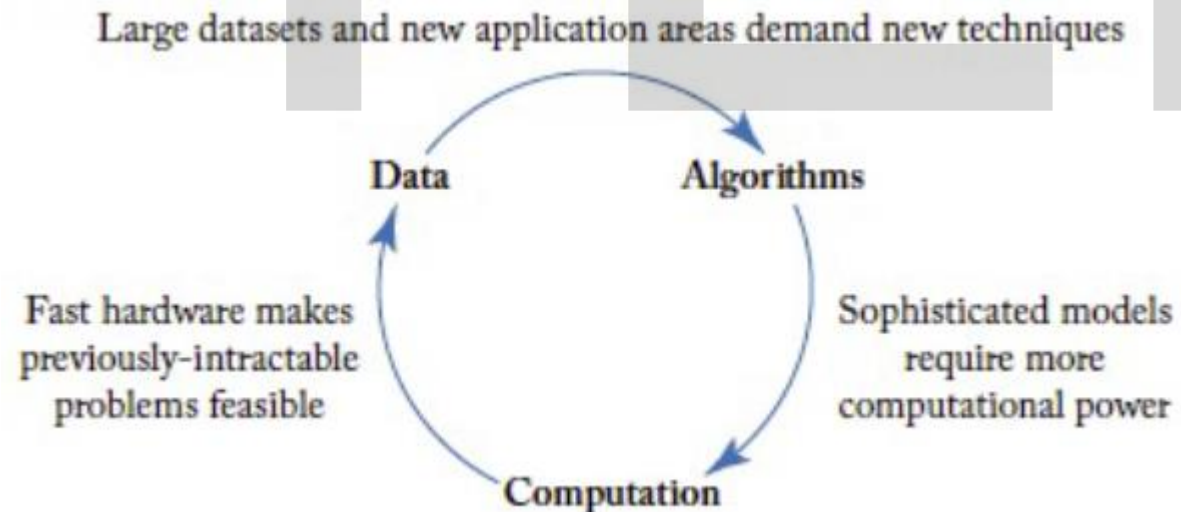


What Actually Changed?

- **Large datasets and new application area demand new techniques**
 - Autonomous driving
- **Sophisticated models require more computational power**
 - LR < RT/DT < DNN < CNN < GNN < LSTM < Transformer
- **Fast hardware makes previously-intractable problems feasible**
 - CPU < GPU < TPU < Accelerators
- **It's a cycle**

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Data Processing

- **Making sense of massive volumes of data**
 - Constrain the tasks
 - 100% correctness is not a necessity (in some domains)
- **Learn from enough examples**
 - Label the data
 - Put GPUs to use
- **Newer domains**
 - No need for data labelling
 - Put accelerators to use

Role of Computing

- **Moore's law**
 - **Observation** that the number of transistors in an integrated circuit doubles about every two years
- **Dennard scaling**
 - As transistors get smaller, their power density stays constant
- **To maintain both**
 - Parallelism is forced on hardware
 - Parallelism is inherent in ML
 - Symbiotic relationship

Role of Computing

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- Who is Moore?
 - Gordon Moore is one of the founders of Intel Corporation

- **Dennard scaling**

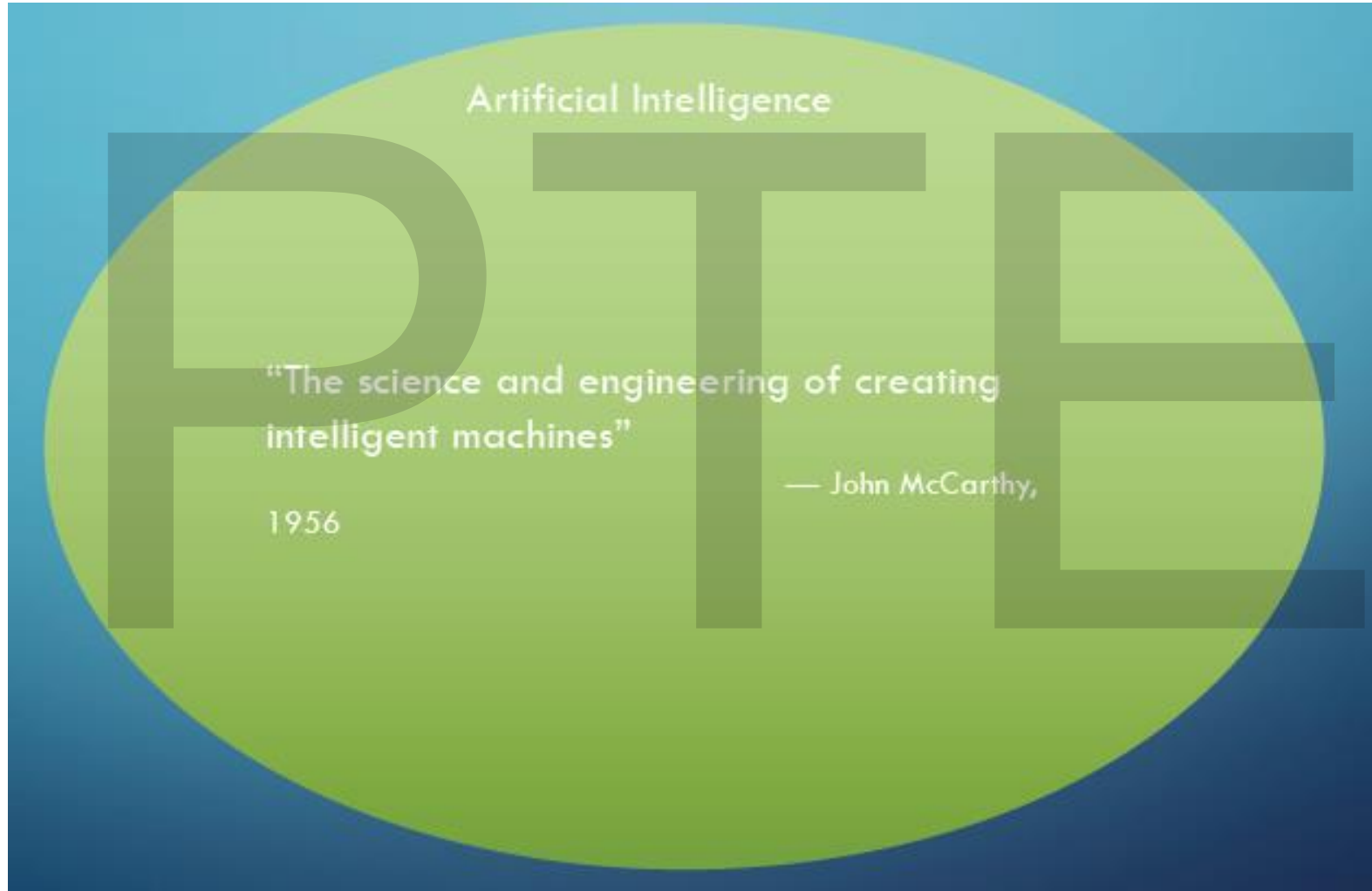
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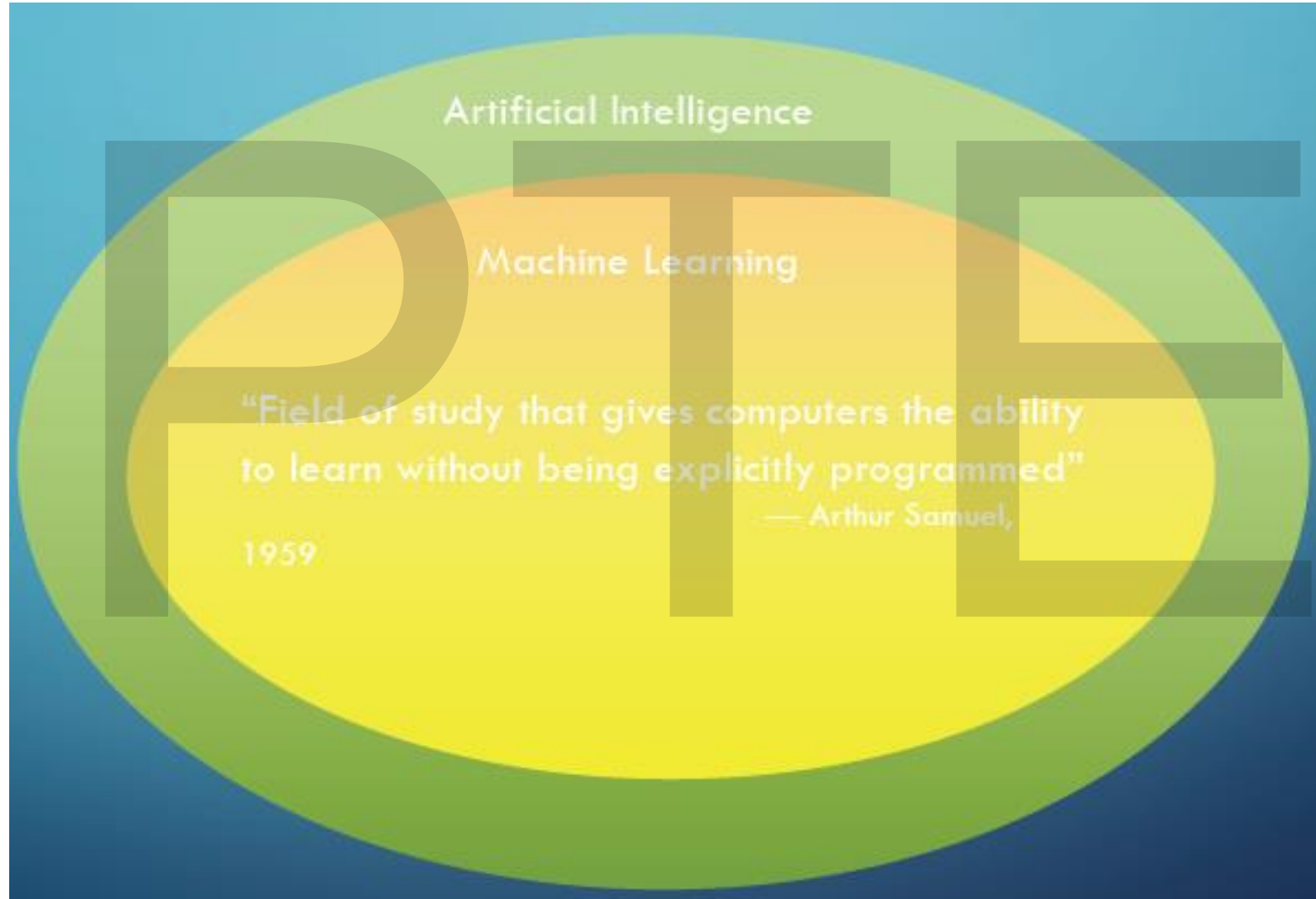
Artificial Intelligence

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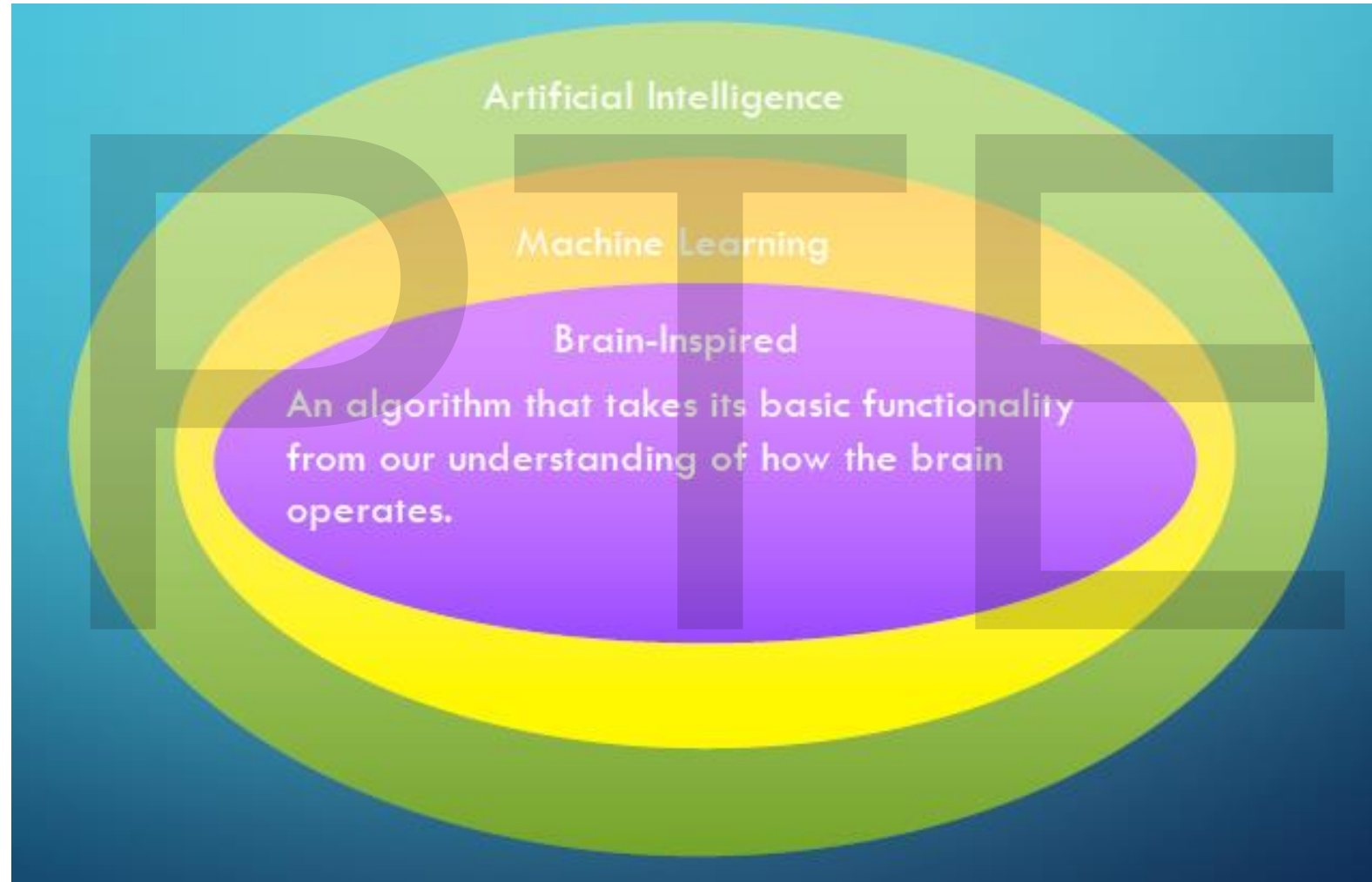


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Machine Learning

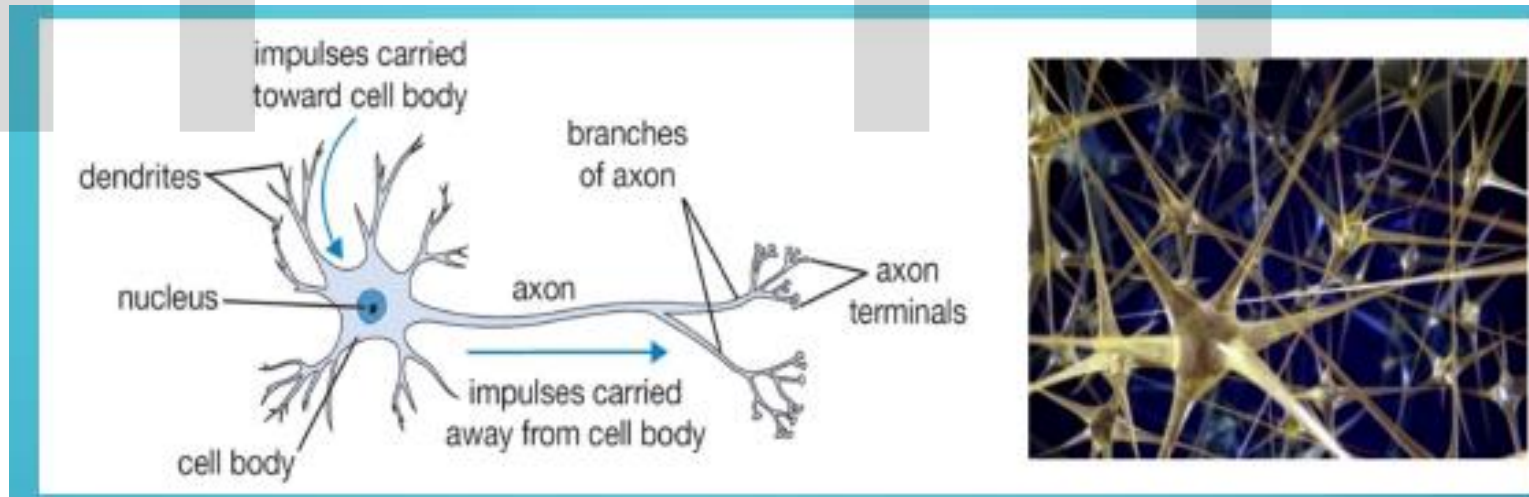


Brain-inspired

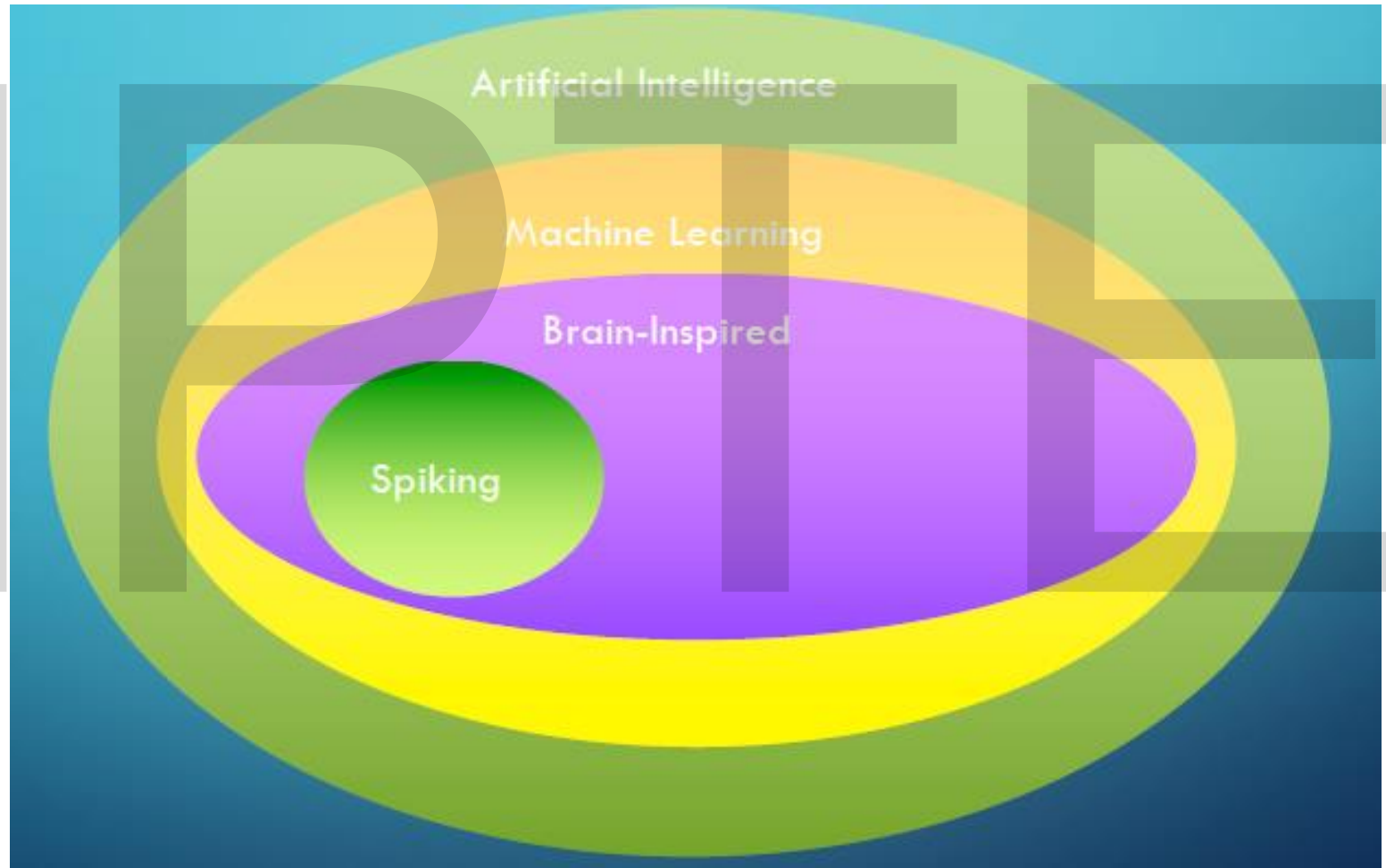


How does the brain work?

- **The basic computational unit of the brain is a neuron**
 - 86B neurons are present in Brain
- **Neurons are connected with nearly 10^{14} – 10^{15} synapses**
- **Neurons receive input signal from dendrites and produce outputs signal along axon, which interact with dendrites of other neurons via synaptic weights**
- **Synaptic weights – learnable and control influence strength**

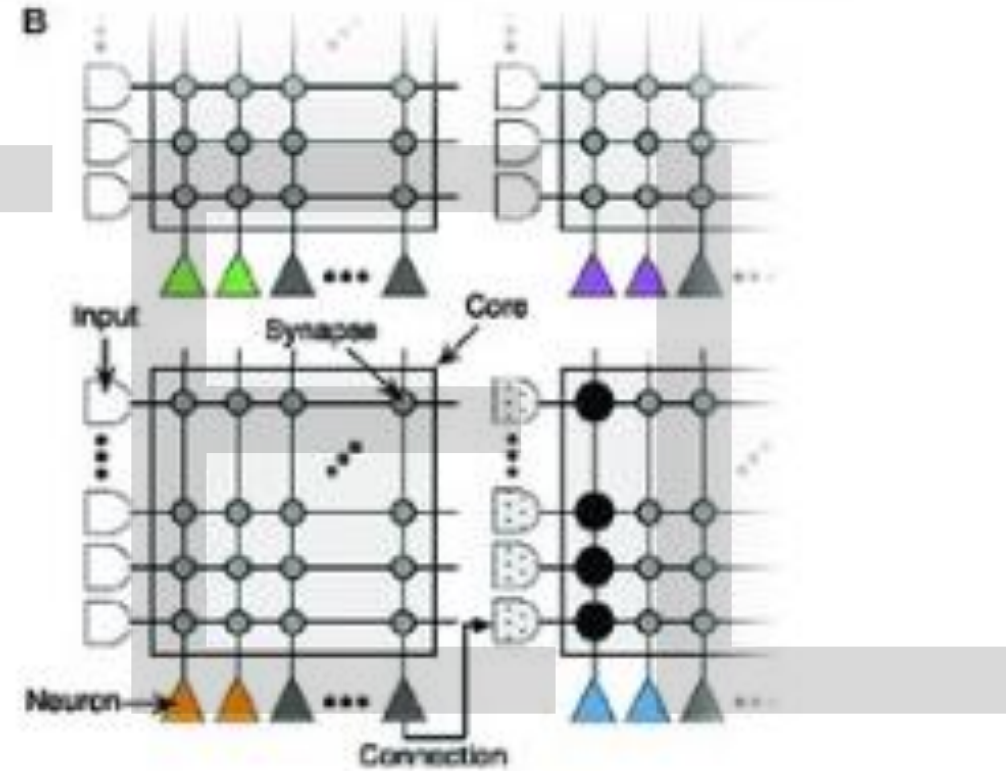


Spiking



Spiking Architecture

- Brain-inspired
- Integrate and fire
- Example: IBM True North
 - 1M neurons
 - 256M synapses
 - 5.4B transistors
 - 73 mW power consumption



TrueNorth Core Architecture